

$$c): yz_x + xz_y = y(2x(\cos(x^2 - y^2))) - x(2y(\cos(x^2 - y^2))) = 0$$

$$d): xz_x - yz_y = xy e^{xy} - yx e^{xy} = 0$$

Exercise # 13.6

Question # 05:

$$f(x, y, z) = 4x^5 y^2 z^3, \quad p(2, -1), \quad u = \frac{1}{3}i + \frac{2}{3}j - \frac{2}{3}k$$

Calculate the partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$ and $f_z(x, y, z)$

$$f_x(x, y, z) = \frac{\partial}{\partial x} (4x^5 y^2 z^3)$$

$$f_x(x, y, z) = 5y^2 z^3 (4x^4)$$

$$f_x(x, y, z) = 20x^4 y^2 z^3$$

$$f_y(x, y, z) = \frac{\partial}{\partial y} (4x^5 y^2 z^3)$$

$$f_y(x, y, z) = 4x^5 z^3 (2y) = 8x^5 y z^3$$

$$f_z(x, y, z) = \frac{\partial}{\partial z} (4x^5 y^2 z^3)$$

$$f_z(x, y, z) = 4x^5 y^2 (3z^2) = 12x^5 y^2 z^2$$

Calculate the gradient of $f(x, y, z)$
 $\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$

$$\nabla f(x, y, z) = 20x^4y^2z^3\mathbf{i} + 8x^5yz^3\mathbf{j} + 12x^5y^2z^2\mathbf{k}$$

evaluate the gradient at the given point
 $P(2, -1, 1)$

$$\nabla f(2, -1, 1) = 20(2)^4(-1)^2(1)^3\mathbf{i} + 8(2)^5(-1)(1)^3\mathbf{j} + 12(2)^5(-1)^2(1)^2\mathbf{k}$$

$$\nabla f(2, -1, 1) = 320\mathbf{i} - 256\mathbf{j} + 384\mathbf{k}$$

Calculate the directional derivative $D_u f(x, y, z)$ at $P(2, -1, 1)$ in the direction of u .

$$D_u f(x, y, z) = \nabla f(x, y, z) \cdot u$$

$$D_u f(2, -1, 1) = \nabla f(2, -1, 1) \cdot u$$

$$D_u f(2, -1, 1) = (320\mathbf{i} - 256\mathbf{j} + 384\mathbf{k}) \cdot \left(\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}\right)$$

Solving the dot product.

$$D_u f(2, -1, 1) = (320)\left(\frac{1}{3}\right) + (-256)\left(\frac{2}{3}\right) + (384)\left(-\frac{2}{3}\right)$$

$$D_u f(2, -1, 1) = \frac{320}{3} - \frac{512}{3} - 256$$

$$\text{Duf}(2, -1, 1) = -320$$

Question No 6

$$f(x, y, z) = ye^{x^2} + z^2, \quad p(0, 2, 3), \quad u = \frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k$$

calculate the partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$ and $f_z(x, y, z)$

$$f_x(x, y, z) = \frac{\partial}{\partial x} (ye^{x^2} + z^2)$$

$$f_x(x, y, z) = yze^{x^2} + 0$$

$$f_x(x, y, z) = yze^{x^2}$$

$$f_y(x, y, z) = \frac{\partial}{\partial y} (ye^{x^2} + z^2)$$

$$f_y(x, y, z) = e^{x^2} + 0$$

$$f_y(x, y, z) = e^{x^2}$$

$$f_z(x, y, z) = \frac{\partial}{\partial z} (ye^{x^2} + z^2)$$

$$f_z(x, y, z) = xye^{x^2} + 2z$$

Calculate the gradient of $f(x, y, z)$

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

$$\nabla f(x, y, z) = yze^{xz}\mathbf{i} + e^{xz}\mathbf{j} + (xye^{xz} + 2z)\mathbf{k}$$

evaluate the gradient at the given point $P(0, 2, 3)$

$$\nabla f(0, 2, 3) = (2)(3)e^{(0)(3)}\mathbf{i} + e^{(0)(3)}\mathbf{j} + ((0)(2)e^{(0)(3)} + 2(3))\mathbf{k}$$

$$\nabla f(0, 2, 3) = 6\mathbf{i} + \mathbf{j} + 6\mathbf{k}$$

Calculate the directional derivative $D_u f(x, y, z)$ at $P(0, 2, 3)$ in the direction of u .

$$D_u f(x, y, z) = \nabla f(x, y, z) \cdot u$$

$$D_u f(0, 2, 3) = \nabla f(0, 2, 3) \cdot u$$

$$D_u f(0, 2, 3) = (6\mathbf{i} + \mathbf{j} + 6\mathbf{k}) \cdot \left(\frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}\right)$$

Solving the dot product.

$$D_u f(0, 2, 3) = (6)\left(\frac{2}{7}\right) + (1)\left(-\frac{3}{7}\right) + (6)\left(\frac{6}{7}\right)$$

$$D_u f(0, 2, 3) = \frac{12}{7} - \frac{3}{7} + \frac{36}{7}$$

$$Duf(2, -1, 1) = \frac{45}{7}$$

Question No 7

$$f(x, y, z) = \ln(x^2 + 2y^2 + 3z^2), P(-1, 2, 4),$$

$$u = -\frac{3}{13}i - \frac{4}{13}j - \frac{12}{3}k$$

calculate the partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$ and $f_z(x, y, z)$

$$f_x(x, y, z) = \frac{\partial}{\partial x} (\ln(x^2 + 2y^2 + 3z^2))$$

$$f_x(x, y, z) = \frac{1}{x^2 + 2y^2 + 3z^2} \cdot \frac{\partial}{\partial x} (x^2 + 2y^2 + 3z^2)$$

$$f_x(x, y, z) = \frac{1}{x^2 + 2y^2 + 3z^2} (2x)$$

$$f_x(x, y, z) = \frac{2x}{x^2 + 2y^2 + 3z^2}$$

$$f_y(x, y, z) = \frac{\partial}{\partial y} (\ln(x^2 + 2y^2 + 3z^2))$$

$$f_y(x, y, z) = \frac{1}{x^2 + 2y^2 + 3z^2} \cdot \frac{\partial}{\partial y} (x^2 + 2y^2 + 3z^2)$$

$$f_y(x, y, z) = \frac{1}{x^2 + 2y^2 + 3z^2} (4y)$$

$$f_y(x, y, z) = \frac{4y}{x^2 + 2y^2 + 3z^2}$$

and,

$$f_z(x, y, z) = \frac{\partial}{\partial z} (\ln(x^2 + 2y^2 + 3z^2))$$

$$f_z(x, y, z) = \frac{1}{x^2 + 2y^2 + 3z^2} \cdot \frac{\partial}{\partial z} (6z)$$

$$f_z(x, y, z) = \frac{6z}{x^2 + 2y^2 + 3z^2}$$

Calculate the gradient of $f(x, y, z)$

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

$$\nabla f(x, y, z) = \frac{2x}{x^2 + 2y^2 + 3z^2}\mathbf{i} + \frac{4y}{x^2 + 2y^2 + 3z^2}\mathbf{j} + \frac{6z}{x^2 + 2y^2 + 3z^2}\mathbf{k}$$

$$\nabla f(x, y, z) = \frac{2}{x^2 + 2y^2 + 3z^2} (x\mathbf{i} + 2y\mathbf{j} + 3z\mathbf{k})$$

evaluate the gradient at the given point $P(1, 2, 4)$

$$\nabla f(1, 2, 4) = \frac{2}{(-1)^2 + 2(2)^2 + 3(4)^2} (-\mathbf{i} + 2(2)\mathbf{j} + 3(4)\mathbf{k})$$

$$\nabla f(1, 2, 4) = \frac{2}{57} (-\mathbf{i} + 4\mathbf{j} + 12\mathbf{k})$$

$$\nabla f(-1, 2, 4) = -\frac{2}{57}i + \frac{8}{57}j + \frac{24}{57}k$$

Calculate the directional derivative $D_u f(x, y, z)$ at $P(-1, 2, 4)$ in the direction of u .

$$D_u f(x, y, z) = \nabla f(x, y, z) \cdot u$$

$$D_u f(-1, 2, 4) = \nabla f(-1, 2, 4) \cdot u$$

$$D_u f(-1, 2, 4) = \left(-\frac{2}{57}i + \frac{8}{57}j + \frac{24}{57}k\right) \cdot \left(-\frac{3}{13}i - \frac{4}{13}j - \frac{12}{13}k\right)$$

Solving the product.

$$D_u f(-1, 2, 4) = \left(-\frac{2}{57}\right)\left(-\frac{3}{13}\right) + \left(\frac{8}{57}\right)\left(-\frac{4}{13}\right) + \left(\frac{24}{57}\right)\left(-\frac{12}{13}\right)$$

$$D_u f(-1, 2, 4) = \frac{2}{247} - \frac{32}{741} - \frac{96}{247}$$

$$D_u f(-1, 2, 4) = -\frac{314}{741}$$

Question No 8:

$$f(x, y, z) = \sin xyz, \quad P\left(\frac{1}{2}, \frac{1}{3}, \pi\right)$$

$$u = \frac{1}{\sqrt{3}}i - \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k$$

Calculate the partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$ and $f_z(x, y, z)$

$$f_x(x, y, z) = \frac{\partial}{\partial x} (\sin xyz)$$

$$f_x(x, y, z) = \cos xyz (yz)$$

$$f_x(x, y, z) = yz \cos xyz$$

$$f_y(x, y, z) = \frac{\partial}{\partial y} (\sin xyz)$$

$$f_y(x, y, z) = \cos xyz (xz)$$

$$f_y(x, y, z) = xz \cos xyz$$

and,

$$f_z(x, y, z) = \frac{\partial}{\partial z} (\sin xyz)$$

$$f_z(x, y, z) = \cos xyz (xy)$$

$$f_2(x, y, z) = xy \cos xyz$$

Calculate the gradient of $f(x, y, z)$

$$\nabla f(x, y, z) = f_x(x, y, z)i + f_y(x, y, z)j + f_z(x, y, z)k$$

$$\nabla f(x, y, z) = yz \cos xyz i + xz \cos xyz j + xy \cos xyz k$$

$$\nabla f(x, y, z) = (yz i + xz j + xy k) \cos xyz$$

evaluate the gradient at the given point $P(\frac{1}{2}, \frac{1}{3}, \pi)$

$$\nabla f(\frac{1}{2}, \frac{1}{3}, \pi) = (\frac{1}{3} \pi i + \frac{1}{2} \pi j + \frac{1}{6} k) \cos(\frac{\pi}{6})$$

$$\nabla f(\frac{1}{2}, \frac{1}{3}, \pi) = \frac{\sqrt{3}}{2} (\frac{1}{3} \pi i + \frac{1}{2} \pi j + \frac{1}{6} k)$$

$$\nabla f(\frac{1}{2}, \frac{1}{3}, \pi) = (\frac{\sqrt{3}}{6} \pi i + \frac{\sqrt{3}}{4} \pi j + \frac{\sqrt{3}}{12} k)$$

Calculate the directional derivative $D_u f(x, y, z)$ at $P(\frac{1}{2}, \frac{1}{3}, \pi)$ in the direction of u .

$$D_u f(x, y, z) = \nabla f(x, y, z) \cdot u$$

$$D_u f\left(\frac{1}{2}, \frac{1}{3}, \pi\right) = \nabla f\left(\frac{1}{2}, \frac{1}{3}, \pi\right) \cdot u$$

$$D_u f\left(\frac{1}{2}, \frac{1}{3}, \pi\right) = \left(\frac{\sqrt{3}\pi}{6}i + \frac{\sqrt{3}\pi}{4}j + \frac{\sqrt{3}}{12}k\right) \cdot$$

$$\left(-\frac{1}{\sqrt{3}}i + -\frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k\right)$$

Solving the product

$$D_u f\left(\frac{1}{2}, \frac{1}{3}, \pi\right) = \left(\frac{\sqrt{3}}{6}\pi\right)\left(\frac{1}{\sqrt{3}}\right) + \left(\frac{\sqrt{3}}{4}\pi\right)$$

$$\left(-\frac{1}{\sqrt{3}}\right) + \left(\frac{\sqrt{3}}{12}\right)\left(\frac{1}{\sqrt{3}}\right)$$

$$D_u f\left(\frac{1}{2}, \frac{1}{3}, \pi\right) = \frac{1}{6}\pi - \frac{1}{4}\pi + \frac{1}{12}$$

$$D_u f\left(\frac{1}{2}, \frac{1}{3}, \pi\right) = \frac{1}{12} - \frac{1}{12}\pi$$

Question NO 15:

$$f(x, y, z) = xy + z^2, \quad P(-3, 0, 4), \quad u = i + j + k$$

$$\nabla f(x, y, z) = yi + xj + 2zk$$

$$\nabla f(-3, 0, 4) = -3j + 8k$$

$$u = (i + j + k)/\sqrt{3}$$

$$D_u f = 5/\sqrt{3}$$

Question No 16

$$f(x, y, z) = y - \sqrt{x^2 + z^2}, \quad P = (-3, 1, 4)$$

$$\nabla f(x, y, z) = -x(x^2 + z^2)^{-1/2} \mathbf{i} + \mathbf{j} - z(x^2 + z^2)^{-1/2} \mathbf{k}$$

$$\nabla f(-3, 1, 4) = (-3/5)\mathbf{i} + \mathbf{j} + (4/5)\mathbf{k}$$

$$u = (2\mathbf{i} - 2\mathbf{j} - \mathbf{k})/3$$

$$D_u f = 0$$

Question No 17

$$f(x, y, z) = \frac{z-x}{z+y}, \quad P(1, 0, -3), \quad q = -6\mathbf{i} + 3\mathbf{j}$$

$$\nabla f(x, y, z) = -\frac{z-x}{(z+y)^2} \mathbf{i} - \frac{z-x}{(z+y)^2} \mathbf{j} + \frac{y+x}{(z+y)^2} \mathbf{k}$$

$$\nabla f(1, 0, -3) = (1/3)\mathbf{i} + (4/9)\mathbf{j} + (1/9)\mathbf{k}$$

$$u = (-16\mathbf{i} + 3\mathbf{j} - 2\mathbf{k})/7$$

$$D_u f = -8/63$$

Question No 18:

$$f(x, y, z) = e^{x+y+3z}, \quad P(-2, 2, -1) \quad q = 2\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$$

$$\nabla f(x, y, z) = e^{x+y+3z}(\mathbf{i} + \mathbf{j} + 3\mathbf{k})$$

$$\nabla f(-2, 2, -1) = e^{-3}(\mathbf{i} + \mathbf{j} + 3\mathbf{k})$$

$$u = (20i - 4j + 5k)/21$$

$$D_u f = (31/21)e^{-3}$$

Directional Derivatives
and gradients

Question NO 41:

$$f(x, y) = 5x^2 + y^4, (4, 2)$$

Calculate the partial derivative $f_x(x, y, z)$
and $f_y(x, y)$

$$f_x(x, y) = \frac{\partial}{\partial x} [5x^2 + y^4]$$

treat y as a constant

$$f_x(x, y) = 10x + 0$$

$$f_x(x, y) = 10x$$

and,

$$f_y(x, y) = \frac{\partial}{\partial y} [5x^2 + y^4]$$

treat x as a constant

$$f_y(x, y) = 0 + 4y^3$$

$$f_y(x, y) = 4y^3$$

The gradient of the function $f(x, y)$ is defined by

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

Substituting the partial derivatives, we obtain

$$\nabla f(x, y) = 10x\mathbf{i} + 4y^3\mathbf{j}$$

the gradient of f at $(4, 2)$ is

$$\nabla f(4, 2) = 10(4)\mathbf{i} + 4(2)^3\mathbf{j}$$

$$\nabla f(4, 2) = 40\mathbf{i} + 32\mathbf{j}$$

Question NO 42:

$$f(x, y) = 5\sin x^2 + \cos 3y, \quad (\sqrt{x}/2, 0)$$

$$\text{Gradient of } f = \nabla f(x, y) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}$$

$$= \frac{\partial}{\partial x} (5\sin x^2 + \cos 3y)\mathbf{i} + \frac{\partial}{\partial y} (5\sin x^2 + \cos 3y)\mathbf{j}$$

$$= 10x\cos x^2\mathbf{i} - 3\sin 3y\mathbf{j}$$

and so,

$$\nabla f\left(\frac{\sqrt{x}}{2}, 0\right) = 10 \times \frac{\sqrt{x}}{2} \times \cos\left(\frac{\sqrt{x}}{2}\right)^2\mathbf{i} - 3\sin(3 \times 0)$$

$$= 5\sqrt{\pi} \times \cos\left(\frac{\pi}{4}\right)i - 3\sin 0j$$

$$= 5\sqrt{\pi} \times \frac{\sqrt{2}}{2}i - 0 = \frac{5}{2}\sqrt{2\pi}$$

Question No 43:

$$f(x, y) = (x^2 + xy)^3; \quad (1, -1)$$

calculate the partial derivatives $f_x(x, y)$ and $f_y(x, y)$

$$f_x(x, y) = \frac{\partial}{\partial x} [(x^2 + xy)^3]$$

treat y as a constant and use the chain rule.

$$f_x(x, y) = 3(x^2 + xy)^{3-1} \frac{\partial}{\partial x} (x^2 + xy)$$

$$f_x(x, y) = 3(x^2 + xy)^2 (2x + y)$$

$$f_x(x, y) = 3(2x + y)(x^2 + xy)^2$$

and,

$$f_y(x, y) = \frac{\partial}{\partial y} [5x^2 + y^4]$$

treat x as a constant and use the chain rule.

$$f_y(x, y) = 3(x^2 + xy)^3 \cdot \frac{\partial}{\partial y} (x^2 + xy)$$

$$f_y(x, y) = 3(x^2 + xy)^2(0 + x)$$

$$f_y(x, y) = 3x(x^2 + xy)^2$$

The gradient of the function $f(x, y)$ is defined by

$$\nabla f(x, y) = f_x(x, y)i + f_y(x, y)j$$

Substituting the partial derivatives, we obtain

$$\nabla f(x, y) = 3(2x + y)(x^2 + xy)^2 i + 3x(x^2 + xy)^2 j$$

$$\nabla f(x, y) = 3(x^2 + xy)^2 [(2x + y)i + xj]$$

the gradient of f at $(-1, -1)$ is

$$\nabla f(-1, -1) = 3((-1)^2 + (-1)(-1))^2 [(2(-1) - 1)i + (-1)j]$$

$$\nabla f(-1, -1) = 3(2)^2 [(-3)i + (-1)j]$$

$$\nabla f(-1, -1) = -12(3i + j)$$

Question No 44

$$f(x,y) = (x^2 + y^2)^{-1/2}, (3,4)$$

calculate the partial derivative f_x
(x,y) and $f_y(x,y)$

$$f_x(x,y) = \frac{\partial}{\partial x} [(x^2 + y^2)^{-1/2}]$$

treat y as a constant and use the chain rule.

$$f_x(x,y) = (-1/2)(x^2 + y^2)^{-1/2-1} \frac{\partial}{\partial x} (x^2 + y^2)$$

$$f_x(x,y) = (-1/2)(x^2 + y^2)^{-3/2} (2x)$$

$$f_x(x,y) = -x(x^2 + y^2)^{-3/2}$$

and,

$$f_y(x,y) = \frac{\partial}{\partial y} [(x^2 + y^2)^{-1/2}]$$

treat x as a constant and use the chain rule

$$f_y(x,y) = (-1/2)(x^2 + y^2)^{-1/2-1} \frac{\partial}{\partial y} (x^2 + y^2)$$

$$f_y(x, y) = (-1/2)(x^2 + y^2)^{-3/2}(2y)$$

$$f_y(x, y) = -y(x^2 + y^2)^{-3/2}$$

The gradient of the function $f(x, y)$ is defined by

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

Substituting the partial derivatives, we obtain

$$\nabla f(x, y) = -x(x^2 + y^2)^{-3/2}\mathbf{i} - y(x^2 + y^2)^{-3/2}\mathbf{j}$$

$$\nabla f(x, y) = (x^2 + y^2)^{-3/2}(-x\mathbf{i} - y\mathbf{j})$$

the gradient of f at $(3, 4)$ is

$$\nabla f(3, 4) = (3^2 + 4^2)^{-3/2}(-3\mathbf{i} - 4\mathbf{j})$$

$$\nabla f(3, 4) = -(25)^{-3/2}(3\mathbf{i} + 4\mathbf{j})$$

$$\nabla f(3, 4) = -1/(125)(3\mathbf{i} + 4\mathbf{j})$$

Question No 45

$$f(x, y, z) = y \ln(x + y + z), (-3, 4, 0)$$

Calculate the partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$ and $f_z(x, y, z)$

$$f_x(x, y, z) = \frac{\partial}{\partial x} [y \ln(x + y + z)]$$

treat y and z are constants

$$f_x(x, y, z) = y \frac{\partial}{\partial x} [\ln(x + y + z)]$$

$$f_x(x, y, z) = y \left(\frac{1}{x + y + z} \right)$$

$$f_x(x, y, z) = \frac{y}{x + y + z}$$

$$f_y(x, y, z) = \frac{\partial}{\partial y} [y \ln(x + y + z)]$$

treat x and z as a constant and use product rule.

$$f_y(x, y, z) = y \frac{\partial}{\partial y} [\ln(x + y + z)] + \ln(x + y + z) \frac{\partial}{\partial y} y$$

$$f_y(x, y, z) = y \left(\frac{1}{x + y + z} \right) + \ln(x + y + z) (1)$$

$$f_y(x, y, z) = \frac{y}{x+y+z} + \ln(x+y+z)$$

and,

$$f_z(x, y, z) = \frac{\partial}{\partial z} [y \ln(x+y+z)]$$

treat x and z as a constant and use product rule.

$$f_z(x, y, z) = \frac{\partial}{\partial z} y \{ \ln(x+y+z) \}$$

treat y and z as constants

$$f_z(x, y, z) = y \frac{\partial}{\partial z} [\ln(x+y+z)]$$

$$f_z(x, y, z) = y \left(\frac{1}{x+y+z} \right)$$

$$f_z(x, y, z) = \frac{y}{x+y+z}$$

The gradient of the function $f(x, y, z)$ is defined by

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

Substituting the partial derivatives, we obtain

$$\nabla f(x, y, z) = \frac{y}{x+y+z} i + \left[\frac{y}{x+y+z} + \ln(x+y+z) \right] j + \frac{y}{x+y+z} k$$

the gradient of f at $(-3, 4, 0)$ is

$$\nabla f(-3, 4, 0) = \frac{4}{-3+4+0} i + \left[\frac{4}{-3+4+0} + \ln(-3+4+0) \right] j + \frac{4}{-3+4+0} k$$

$$\nabla f(-3, 4, 0) = 4i + 4j + 4k$$

Question No 46:

$$f(x, y, z) = y^2 z \tan^3 x \quad (\pi/4, -3, 1)$$

calculate the partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$ and $f_z(x, y, z)$

$$f_x(x, y, z) = \frac{\partial}{\partial x} [y^2 z \tan^3 x]$$

treat y and z as constants

$$f_x(x, y, z) = y^2 z \frac{\partial}{\partial x} [\tan^3 x]$$

$$f_x(x, y, z) = y^2 z (3 \tan^2 x) (\sec^2 x)$$

$$f_x(x, y, z) = 3 z \sec^2 x \tan^2 x y^2$$

$$f_y(x, y, z) = \frac{\partial}{\partial y} [y^2 z \tan^3 x]$$

treat x and z as a constant and use product rule.

$$f_y(x, y, z) = z \tan^3 x \frac{\partial}{\partial y} [y^2]$$

$$f_y(x, y, z) = 2 y z \tan^3 x$$

and,

$$f_z(x, y, z) = \frac{\partial}{\partial z} [y^2 \tan^3 x z]$$

treat x and y as constants

$$f_z(x, y, z) = y^2 \tan^3 x \frac{\partial}{\partial z} [z]$$

$$f_z(x, y, z) = y^2 \tan^3 x$$

The gradient of the function
 $f(x, y, z)$

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

Substituting the partial derivatives we obtain

$$\nabla f(x, y, z) = 3z \sec^2 x \tan^2 xy^2 \mathbf{i} + 2yz \tan^3 x \mathbf{j} + y^2 \tan^3 x \mathbf{k}$$

the gradient of f at $(\pi/4, -3, 1)$ is

$$\begin{aligned} \nabla f(\pi/4, -3, 1) &= 3(1)(-3)^2 \sec^2\left(\frac{\pi}{4}\right) \tan^2\left(\frac{\pi}{4}\right) \mathbf{i} \\ &+ 2(-3)(1) \tan^3\left(\frac{\pi}{4}\right) \mathbf{j} + (-3)^2 \tan^3\left(\frac{\pi}{4}\right) \mathbf{k} \end{aligned}$$

$$\nabla f(\pi/4, -3, 1) = 54\mathbf{i} - 6\mathbf{j} + 9\mathbf{k}$$